

Junlong Ning

✉ junlongn@andrew.cmu.edu | 📞 (+86) 183-2330-6458 | 📍 Pittsburgh, PA
🐙 GitHub | 🔗 LinkedIn | 🎓 Graduating Dec 2027

EDUCATION

Carnegie Mellon University (CMU)
M.S. in Artificial Intelligence Engineering (AIE)

Pittsburgh, PA
Expected Dec 2027

Chongqing University (CQU)
B.E. in Robotics Engineering

Chongqing, China
Sep 2022 – Jun 2026

SKILLS

Frameworks & ML Methods: LangGraph (ReAct/StateGraph), PyTorch, HuggingFace Transformers, DeepSpeed (ZeRO-2), PEFT/LoRA, Conformal Prediction, Multimodal Fusion, Uncertainty Quantification

Agent Infra: FastAPI, Redis, Docker, CI/CD (GitHub Actions), LangSmith, SHAP, RAG (ChromaDB, sentence-transformers)

Tools, Languages & Systems: Git/Linux, Python, C++/C, MATLAB, OpenCV, ROS/ROS2, NVIDIA Jetson NX

INTERNSHIP

Innovation AI

May 2026 – Aug 2026

Machine Learning Engineer Intern

- Built a **ReAct agent** with **LangGraph StateGraph** orchestrating a **5-tool graph** (price, sentiment, quant, retrieval, memory), with conditional error-fallback edges and **LangSmith** tracing over every Thought→Action→Observation transition.
- Designed two agent-callable tools behind a typed JSON **function-calling schema**: a FinBERT sentiment module and an XGBoost/LightGBM predictor wrapped with **SHAP** explainability for human-readable justifications.
- Implemented a **RAGTool** (sentence-transformers + ChromaDB) invoked conditionally by the agent's own policy, plus a SQLite **MemoryStore** enabling multi-turn, session-aware queries where the agent recalls and re-evaluates its own past decisions.
- Deployed the agent behind a **FastAPI + SSE** streaming backend with **Redis**-backed response caching and rate limiting, containerized with Docker and shipped via GitHub Actions CI/CD.
- Built an evaluation harness scoring tool-calling accuracy, latency, and groundedness on a 15-ticker benchmark (**93%** groundedness, avg. **13.6s** latency), catching the agent silently diluting a QuantTool SELL signal into HOLD.

RESEARCH

When2Ask: Uncertainty-Aware Interactive Navigation for VLMs in Urban Scenes

Oct 2024 – Mar 2025

Co-First Author & Research Assistant, New York University · Advised by Prof. Chen Feng (NYU)

- Designed an **autonomous navigation agent** that treats "ask a human" as a first-class action, escalated only under high uncertainty via a multi-step **Conformal Prediction (CP)** module — reducing over-reliance on human input.
- Built the agent's perception layer with **CLIP-based image-language scoring** to evaluate candidate directions, feeding confidence estimates directly into the CP-gated action-selection policy.
- Built a calibration pipeline on 500 street-view samples across 5 cities to derive per-city CP confidence thresholds ($\epsilon=0.8$), delivering an average **+25 pp Success Rate** gain over Relevance/Entropy baselines.
- Deployed the agent's 4-option prediction-set action space on V-IRL; best-city result **76.9% SR in New York** (+30.7 pp vs. Relevance) with lower Normalized Time Steps, improving both accuracy and step-efficiency.

View2Verify: Forged Image Detection, Localization & Interpretation via VLMs

Sep 2025 – May 2026

Undergraduate Thesis, Chongqing University · Advised by Assoc. Prof. Xiangpeng Li

- Proposed **View2Verify**, a dual-stream multimodal framework fusing a semantic stream (CLIP-ViT) with a frequency-domain stream (SRM high-pass filter), jointly improving detection accuracy and interpretability for deepfake localization.
- Designed a **frequency-guided Attention Mask Generation (AMG)** module that produces pixel-level spatial masks from SRM-extracted high-frequency noise residuals, directing the semantic branch to suspected forgery regions.
- Applied **LoRA** parameter-efficient fine-tuning ($r=128, \alpha=256$) to LLaVA-1.5-7B with **DeepSpeed ZeRO-2** under a 24 GB single-GPU constraint, markedly reducing training overhead while preserving reasoning capability.
- Achieved **87.16% accuracy** ($F1=0.9225$, language detail score 88.6%) on FaceForensics++ (FakeClue subset); 71.68% on cross-domain READFake — outperforming LLaVA-1.5, InternVL2-8B, and DeepSeek-VL2-Tiny baselines.

Spatial Performance Evaluation of MLLMs in Outdoor Street Scenes

Jan 2025 – Mar 2025

Research Assistant, New York University · Advised by Prof. Saining Xie & Prof. Chen Feng (NYU)

- Designed a standardized evaluation framework benchmarking spatial-understanding capabilities of frontier MLLMs (GPT-4V, Gemini, Claude); built an automated Python pipeline identifying key failure modes for embodied-agent development.

PUBLICATIONS

J. Ning, Y. Li, et al., "When2Ask: Uncertainty-aware interactive navigation in urban scenes." *Manuscript in preparation.*

J. Ning, "Biomimetic Microrobots for Navigation in Viscous Fluids: Inspired by Bacterial Locomotion." *IEEE ICARA 2025.*